

5.4 给定速度场 $u=x^2y+y^2, v=x^2-y^2x, w=0$, 问: (1) 是否存在不可压缩流函数和速度势函数? (2) 如存在, 请给出它们的具体形式; (3) 写出微团变形速率张量各分量及刚体旋转角速度.

步骤	计算式	结论
1. 连续方程验证	$\frac{\partial u}{\partial x} = 2xy, \frac{\partial v}{\partial y} = -2xy, \frac{\partial w}{\partial z} = 0$ $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$	不可压缩流函数存在
2. 无旋条件验证	$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$ $\frac{\partial v}{\partial x} = 2x - y^2, \frac{\partial u}{\partial y} = x^2 + 2y$ $\omega_z = \frac{1}{2} (-x^2 - y^2 + 2x - 2y) \neq 0$	速度势函数不存在

(2) 求不可压缩流函数 $\psi(x, y)$

步骤	计算式
1. 对y积分 $u = \frac{\partial \psi}{\partial y}$	$\psi = \int (x^2y + y^2) dy = \frac{1}{2}x^2y^2 + \frac{1}{3}y^3 + f(x)$
2. 对x求偏导结合 $v = -\frac{\partial \psi}{\partial x}$	$-\frac{\partial \psi}{\partial x} = -xy^2 - f'(x) = x^2 - y^2x$ $f'(x) = -x^2$
3. 对x积分 $f'(x)$	$f(x) = -\frac{1}{3}x^3 + C \text{ (取 } C=0 \text{)}$
4. 流函数形式	$\psi = \frac{1}{2}x^2y^2 + \frac{1}{3}y^3 - \frac{1}{3}x^3$

(3) 微团变形速率张量与刚体旋转角速度

物理量	分量计算式
变形速率张量 S_{ij}	$S_{xx} = \frac{\partial u}{\partial x} = 2xy$
	$S_{yy} = \frac{\partial v}{\partial y} = -2xy$
	$S_{zz} = \frac{\partial w}{\partial z} = 0$
	$S_{xy} = S_{yx} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) = \frac{1}{2} (x^2 + 2y + 2x - y^2)$
	$S_{xz} = S_{zx} = \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) = 0$
刚体旋转角速度 $\vec{\omega}$	$S_{yz} = S_{zy} = \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) = 0$
	$\omega_x = \frac{1}{2} \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial z} \right) = 0$
	$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial v}{\partial x} \right) = 0$
	$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (-x^2 - y^2 + 2x - 2y)$
	$\vec{\omega} = \omega_z \vec{k}$

5.6 不可压无界流场中有一对等强度 Γ 的点涡, 方向相反, 分别放在 $(0, h)$ 和 $(0, -h)$ 点上. 无穷远处有一股来流 U_∞ , 恰好使这两个涡停留不动, 求流线方程.

1. 流函数叠加

$$\psi_1 = -\frac{\Gamma}{2\pi} \ln \sqrt{x^2 + (y-h)^2}, \quad \psi_2 = \frac{\Gamma}{2\pi} \ln \sqrt{x^2 + (y+h)^2}, \quad \psi_3 = -U_\infty y$$

$$\psi_{\text{总}} = \frac{\Gamma}{4\pi} \ln \frac{x^2 + (y+h)^2}{x^2 + (y-h)^2} - U_\infty y$$

2. 确定 U_∞

$$V_{\text{诱导}} = \frac{\Gamma}{4\pi h} = U_\infty$$

3. 总流函数

$$\psi_{\text{总}} = \frac{\Gamma}{4\pi} \left(\ln \frac{x^2 + (y+h)^2}{x^2 + (y-h)^2} - \frac{y}{h} \right)$$

4. 流线方程

$$\ln \frac{x^2 + (y+h)^2}{x^2 + (y-h)^2} - \frac{y}{h} = \text{const}$$

不可压无旋流“点涡+均匀来流”叠加问题解题模板

步骤	操作内容	公式/计算
1	确定各流动的流函数	- 均匀来流 (U_∞ 沿x轴): $\psi_1 = \pm U_\infty y$ (符号依来流方向定) - 点涡 (强度 Γ , 位置 (x_0, y_0)): $\psi_{\text{涡}} = \frac{\Gamma}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-y_0)^2}$ (逆时针 Γ 为正)
2	流函数叠加	$\psi_{\text{总}} = \psi_{\text{来流}} + \sum \psi_{\text{涡}}$
3	利用“涡停留”条件求来流速度	某涡处诱导速度 (其他涡+来流) 为0: $U_\infty = \sum \frac{\Gamma_i}{2\pi r_i} \text{ (} r_i \text{为涡间距, 速度分量需匹配方向)}$
4	代入速度关系简化流函数	将 U_∞ 与 Γ 的关系代入 $\psi_{\text{总}}$
5	写流线方程	令 $\psi_{\text{总}} = \text{常数}$, 整理得流线表达式

均匀来流	$\varphi = xU_\infty \cos \alpha + yU_\infty \sin \alpha$ $\psi = xU_\infty$	$\psi = yU_\infty \cos \alpha - xU_\infty \sin \alpha$ $\psi = yU_\infty$	$\bar{V} = U_\infty \cos \alpha \vec{i} + U_\infty \sin \alpha \vec{j}$ $u = U_\infty, v = 0$
平面点源 $Q>0$ 平面点汇 $Q<0$	$\varphi = \frac{Q}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-y_0)^2}$ $\varphi = \frac{Q}{2\pi} \ln \sqrt{x^2 + y^2} = \frac{Q}{2\pi} \ln r$	$\psi = \frac{Q}{2\pi} \arctg \frac{y-y_0}{x-x_0}$ $\psi = -\frac{\Gamma}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-y_0)^2}$	$u = \frac{Q(x-x_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $v = \frac{Q(y-y_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $V_r = \frac{Q}{2\pi r}, V_\theta = 0$ $u = -\frac{\Gamma(y-y_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $v = -\frac{\Gamma(x-x_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $V_\theta = \frac{\Gamma}{2\pi r}, V_r = 0$
平面点涡	$\varphi = \frac{\Gamma}{2\pi} \theta$	$\psi = -\frac{\Gamma}{2\pi} \ln r$	$u = -\frac{\Gamma(y-y_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $v = -\frac{\Gamma(x-x_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $V_\theta = \frac{\Gamma}{2\pi r}, V_r = 0$
无环量圆柱绕流	$\varphi = \frac{m}{2\pi} \frac{x-x_0}{(x-x_0)^2 + (y-y_0)^2}$ $\varphi = \frac{m}{2\pi} \frac{x}{x^2 + y^2} = \frac{m \cos \theta}{2\pi r}$	$\psi = -\frac{m}{2\pi} \frac{y-y_0}{(x-x_0)^2 + (y-y_0)^2}$ $\psi = -\frac{m}{2\pi} \frac{y}{x^2 + y^2} = -\frac{m \sin \theta}{2\pi r}$	$V_r = -\frac{m \cos \theta}{2\pi r^2}, V_\theta = -\frac{m \sin \theta}{2\pi r^2}$ $V_r = U_\infty \left(1 - \frac{a^2}{r^2}\right) \cos \theta, V_\theta = -U_\infty \left(1 + \frac{a^2}{r^2}\right) \sin \theta$
有环量圆柱绕流 逆时针 $\Gamma>0$	$\varphi = U_\infty \left(r + \frac{a^2}{r}\right) \cos \theta + \frac{\Gamma}{2\pi} \theta$	$\psi = U_\infty \left(r - \frac{a^2}{r}\right) \sin \theta - \frac{\Gamma}{2\pi} \ln r$	$V_r = U_\infty \cos \theta + \frac{U_\infty a^2}{r^3}, V_\theta = -U_\infty \sin \theta$ $V_r = U_\infty \left(1 - \frac{a^2}{R^2}\right) \cos \theta, V_\theta = -U_\infty \left(1 + \frac{a^2}{2R^2}\right) \sin \theta$
平面钝体绕流	$\varphi = U_\infty r \cos \theta + U_\infty b \ln r$	$\psi = U_\infty r \sin \theta + U_\infty b \theta$	$V_r = U_\infty \cos \theta + \frac{U_\infty b}{r}, V_\theta = -U_\infty \sin \theta$
圆球绕流	$\varphi = U_\infty \left(R + \frac{a^3}{2R^2}\right) \cos \theta$	$\psi = -\frac{1}{2} U_\infty \left(R^2 - \frac{a^3}{R}\right) \sin^2 \theta$	$V_r = U_\infty \left(1 - \frac{a^3}{R^2}\right) \cos \theta, V_\theta = -U_\infty \left(1 + \frac{a^3}{2R^2}\right) \sin \theta$

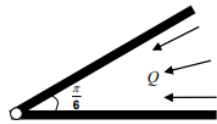
二. “均匀流+点源+点汇(轴对称)”解题流程模板

步骤	操作内容	公式/计算 (柱坐标 $z-r$)
1	确定各流动的流函数	- 均匀流 (U_∞ 沿z轴): $\psi_1 = \frac{1}{2} U_\infty r^2$ - 点源 (强度 Q , 位置 $z=a$): $\psi_2 = -\frac{Q}{4\pi} \frac{z-a}{\sqrt{(z-a)^2 + r^2}}$ - 点汇 (强度 Q , 位置 $z=-a$): $\psi_3 = \frac{Q}{4\pi} \frac{z+a}{\sqrt{(z+a)^2 + r^2}}$
2	流函数叠加	$\psi_{\text{总}} = \psi_1 + \psi_2 + \psi_3$
3	物面方程 ($\psi_{\text{总}}=0$)	令 $\psi_{\text{总}}=0$, 整理得物面的 $z-r$ 关系
4	代入特征参数化简	引入参数 (如 $b^2 = \frac{Q}{2\pi U_\infty}$), 消去 Q 或 U_∞
5	边界条件求解 (如 $z=0, r=h$)	将特殊点代入物面方程, 得到几何参数关系
6	驻点条件 ($V=0$)	叠加各流动的速度分量, 令总速度为0, 求解驻点位置

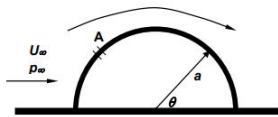
● 例题

1. 30度墙角地漏, 看做平面点汇, 其速度势和速度场

$$\varphi = -\frac{Q}{\pi} \ln r = -\frac{6Q}{\pi} \ln r$$
$$V_r = \frac{\partial \varphi}{\partial r} = -\frac{6Q}{\pi r}$$



已知无穷远来流的风速为 U_∞ , 压力为 p_∞ , 绕过一个半径为 a , 长度为 L 的半圆柱型帐篷, 设内部压力为 p_i , 用无粘理论推导由于内外压差帐篷受到的升力的表达式. 如果风很大, 在帐篷表面开个孔A, 使得内压和该处表面压力相等, 问A孔对应的 $\theta=?$ 使得帐篷受到的升力为零. (若是半球如何?)



$$p_{r=a} = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$$

$$F_y = p_i 2a + \iint [-p_{r=a} dA \sin \theta] = 2p_i a - \int_0^\pi [p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)] \sin \theta a d\theta$$

$$F_y = 2ap_i - 2ap_\infty - \frac{5}{3} \rho a U_\infty^2$$

若开孔后升力为0, $p_i = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$, 代入 $F_y=0$,

则 $\rho a U_\infty^2 (1 - 4 \sin^2 \theta) + \frac{5}{3} \rho a U_\infty^2 = 0, \therefore \sin^2 \theta = \frac{2}{3}, \theta = 54.7^\circ, 125.3^\circ$

● 如果是半球体

$$p_{r=a} = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta)$$

$$F_y = p_i \pi a^2 + \iint [-p_{r=a} \sin \theta \cos \theta dA] = \pi a^2 \sin \theta \cos \theta d\theta, \theta=0 \sim \pi, \cos \theta=2(0 \sim \frac{\pi}{2})$$

$$F_y = \pi a^2 \rho U_\infty^2 \left[\frac{1}{2} \left(\frac{1}{2} + \frac{1}{2} \right) \right] = \pi a^2 \rho U_\infty^2 \left(1 - \frac{9}{4} \sin^2 \theta \right) \sin \theta \cos \theta d\theta$$

$$= \pi a^2 \rho U_\infty^2 \left(p_\infty + \frac{1}{2} \rho U_\infty^2 \right) + \rho U_\infty^2 \frac{9}{4} a^2 \int_0^{\frac{\pi}{2}} \cos \theta \sin^3 \theta d\theta$$

$$= \pi a^2 \rho U_\infty^2 \left(p_\infty + \frac{1}{2} \rho U_\infty^2 \right) + \rho U_\infty^2 \frac{9}{4} a^2 \frac{3\pi}{8}$$

$$= \pi a^2 \rho U_\infty^2 p_\infty + \frac{11}{32} \pi a^2 \rho U_\infty^2$$

若开孔后升力为0, $p_i = p_\infty + \frac{1}{2} \rho U_\infty^2 (1 - \frac{9}{4} \sin^2 \theta)$, 代入 $F_y=0$,

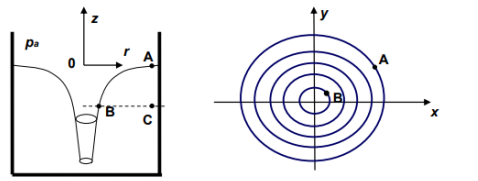
则 $\frac{\pi a^2 \rho U_\infty^2}{2} (1 - \frac{9}{4} \sin^2 \theta) + \frac{11}{32} \pi a^2 \rho U_\infty^2 = 0, \therefore \sin^2 \theta = \frac{3}{4}, \theta = 60^\circ, 120^\circ$

在 $\theta=60^\circ, 120^\circ$ 开孔升力为0.

● 例题

海底漩涡, 在远离中心的速度分布可以近似看成是一个平面点涡 $\varphi = \frac{\Gamma}{2\pi} \theta, \psi = -\frac{\Gamma}{2\pi} \ln r$, 要求

确定自由面的形状.

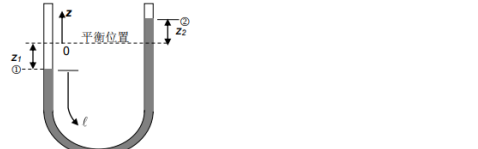


$$V_\theta = \frac{1}{r} \frac{\partial \varphi}{\partial \theta} = \frac{\Gamma}{2\pi r}, \text{ 自由面上任选B远离小孔, A处 } r \rightarrow \infty, \text{ 速度为零, 静力学}$$

$$\frac{p_0}{\rho} + \frac{1}{2} V_A^2 = \frac{p_0}{\rho} + \frac{1}{2} V_B^2 - g z_B, z_B = -z, \frac{1}{2} \frac{r^2}{4\pi^2 r^4} + g z = 0$$

$$\therefore z = \frac{r^2}{8g\pi^2 r^2}$$

1. 等截面开口U型管内总长L的水柱, 因初始扰动水柱偏离平衡位置液面差为 h_0 , 开始在管内振荡, 不计粘性, 求振荡周期, 振荡规律和振荡速度.



解: 无粘, 不可压, 一维流动, 建立坐标如图, 某t时刻对两端的液面有
$$\int_1^2 \frac{\partial \bar{V}}{\partial t} \cdot d\vec{l} + \left(\frac{V^2}{2} + \frac{p}{\rho} + g z \right)_2 = \left(\frac{V^2}{2} + \frac{p}{\rho} + g z \right)_1$$

不可压, 所以 $V_1 = V_2 = V(t) = \frac{dz}{dt}, p_1 = p_2 = p_0$

$$L \frac{d^2 z}{dt^2} + g z = g z, z_1 = -z_2$$

$$L \frac{d^2 z}{dt^2} + 2g z = 0$$

$$\frac{d^2 z}{dt^2} + \frac{2g}{L} z = 0, (r^2 + \frac{2g}{L} = 0, r = \pm i \sqrt{\frac{2g}{L}})$$

$$z_2 = c_1 \cos \left(\sqrt{\frac{2g}{L}} t \right) + c_2 \sin \left(\sqrt{\frac{2g}{L}} t \right)$$

$$r_1 = r_2, y = (c_1 + c_2 x) e^{ix}$$

$$r = c \pm di, y = e^{ax} (c_1 \cos dx + c_2 \sin dx)$$

$$t=0: z_2 = \frac{h_0}{2}, V_2 = \frac{dz_2}{dt} = 0$$

$$\therefore c_1 = \frac{h_0}{2}, c_2 = 0$$

$$z_2 = \frac{h_0}{2} \cos \sqrt{\frac{2g}{L}} t, \text{ 振荡周期}$$

$$T = \frac{2\pi}{\sqrt{\frac{2g}{L}}} = 2\pi \sqrt{\frac{L}{2g}}$$